Analysis Guiding Questions

When using the guiding questions below, consider that you are taking on the role of the XAI system *designer* and the questions address characteristics of the *end-users* and the *context* in which the system will be used.

Personas

Impact Factor(s)	Guiding Question	Situation Illustration
Decision-maker's domain expertise	What do I have learned about your expertise and knowledge in the decision-making domain? Are you and expert or novice?	Experts have more knowledge in the decision-making domain and tend to question more the AI recommendation. Novices tend more to follow AI advice.
Task familiarity	What is your familiarity with the task I am designing for?	When a decision-maker has high familiarity with a decision-making task, he/she will know better what to expect from the AI recommendation, different from someone with low familiarity.
AI literacy	What do I have learned about your knowledge on artificial intelligence systems? Are you a lay user or do you understand fundamentals on how these systems work?	A user with knowledge on AI will be able to expect the potential behavior of the system and question its output. A lay user tends to follow blindly the AI advice.
AI familiarity	How familiar are you with the use of artificial intelligence systems? Have you interacted with them in a regular basis or is it your first experience?	A user without experience interacting with AI system will be more cautious to accept its advice, different from a user familiar with these systems.
Openness to a new experience	How open are you to try a new experience? And to interact with a new system? Enthusiast, open-minded, or skeptic?	When a user is enthusiast or has an open mind to interact with a new system, he/she will be more open to accept an AI advice, different from skeptical users.
Detect and correct AI's mistakes	What do I have learned about your ability to examine the correctness of a recommendation?	The decision-maker's characteristics will frame his/her ability to detect AI mistakes. Someone without this ability can follow blindly AI advice even when it is wrong.

Interest in the task	How interested are you in the decision-making task? Are you cautious when following an advice for making a decision or you not show close attention to it?	A decision-maker who has interest in the current task will be more cautious when following AI advice, different from someone without interest who won't show close attention.
Decision-maker's autonomy	How motivated are you to keep your interests and values when making a decision?	A decision-maker with high autonomy will ensure his/her own opinion when making a decision, different from someone with low autonomy who tends to follows more the AI advice.
Algorithm aversion	How averse are you to interact with an artificial intelligence agent? Do you discount a recommendation because it was made by a computing system? Do you appreciate it when you have a positive first impression of the system?	A user with aversion to use computing systems will reject more an AI advice, different from someone without this aversion.
Decision-maker's self-confidence	What do I have learned about your own confidence for making a decision in the context I am designing for? Do you think to be able to achieve better decisions by yourself instead of using the artificial intelligence recommendations?	A decision-maker with high self-confidence will pay close attention to his/her own decision compared to the AI advice, different from someone with low self-confidence.
Decision-maker's cognitive load	What do I have learned about your ability to handle excessive information? Do you tend to skip information or analyze each one in detail when making a decision?	A decision-maker who tends to skip information will accept more the AI advice, different from someone who analyze information in detail and based on that can identify AI mistakes.

Scenarios of Use

Impact Factor(s)	Guiding Question	Situation Illustration
Decision-making context	How is characterized the decision-making context I am designing for? High-risk or low-risk context?	Wrong decisions made in high-risk context, such as medicine, can have greater consequences to the individuals involved, different from low-risk context, such as leisure.

Degree of vulnerability	How vulnerable and uncertain are the situations belonging to this context?	In certain contexts, such as medicine and finance, the decisions made can generate vulnerable situations to the individuals involved. Not all contexts have vulnerability.
Time pressure for decision-making	How urgent decisions must me made in this context? Why?	When decisions have be made under a high time pressure, the decision-maker can accept more the AI advice compared when there is low or no time pressure.
Domain complexity	How complex is the decision-making domain according to what I have learned about your [user] expertise?	A high complex domain can lead the decision-maker to follow more the AI advice due to its ability to analyze several kinds of information.
Task complexity	What is the decision-making task I am design for? How difficulty will be for you to make a decision in this task?	A high difficulty task can also led novice decision-makers to follow blindly the AI advice, compared to low difficulty tasks and when the decision-maker is expert.